RDP remote control

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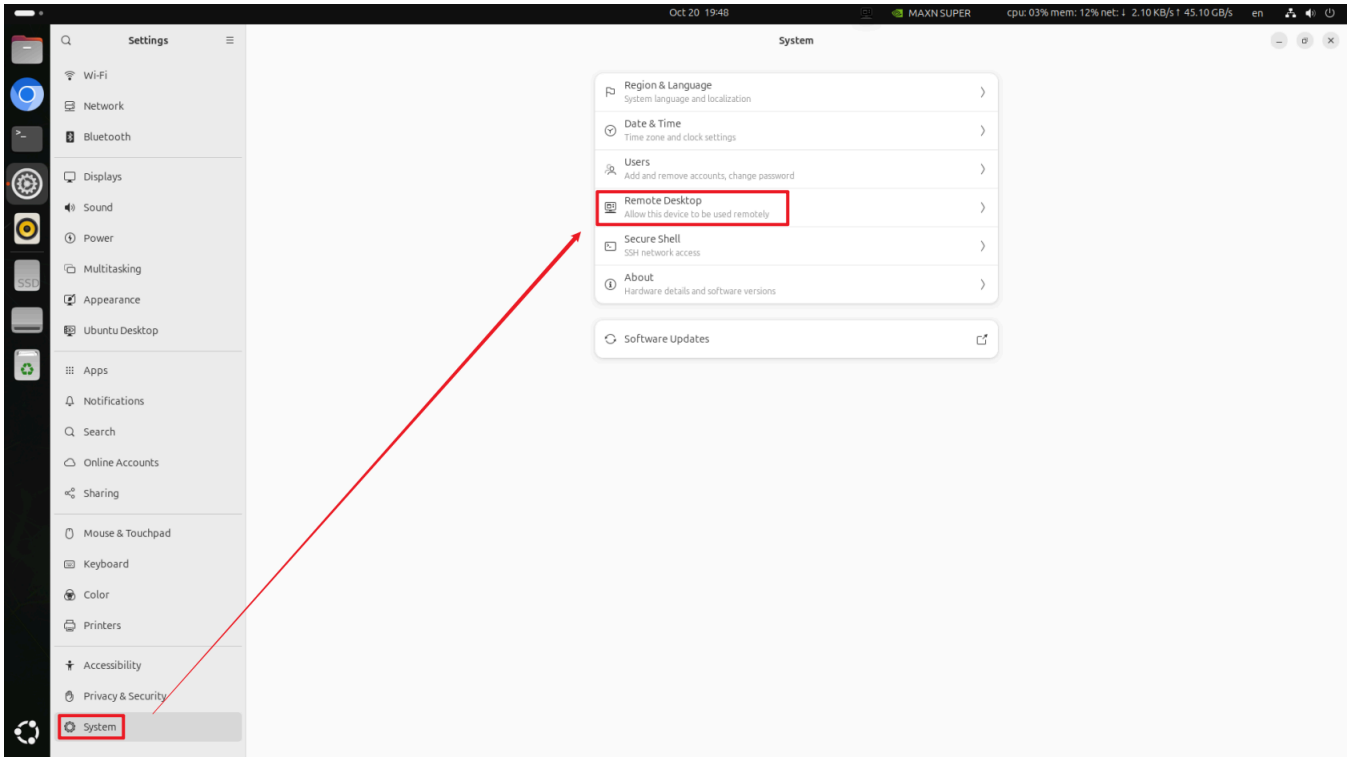
This tutorial introduces logging in to the Jetson Ubuntu desktop remotely from Windows via RDP. The current device uses the system-level Remote Login of GNOME Remote Desktop that ships with Ubuntu 24.04, and does not use `xrdp`.

1. Requirements

- The Jetson is connected to the network and the Windows computer can reach the IP address of the Jetson; for the initial setup it is recommended to use the same LAN.
- GNOME Remote Desktop is installed and enabled on the Jetson; RDP uses the TCP `3389` and `3390` ports.
- Remote login uses the Linux user name of the Jetson and the system login password.
- Before switching the desktop mode or restarting the graphical login manager, make sure that a fallback management entry such as SSH is available.

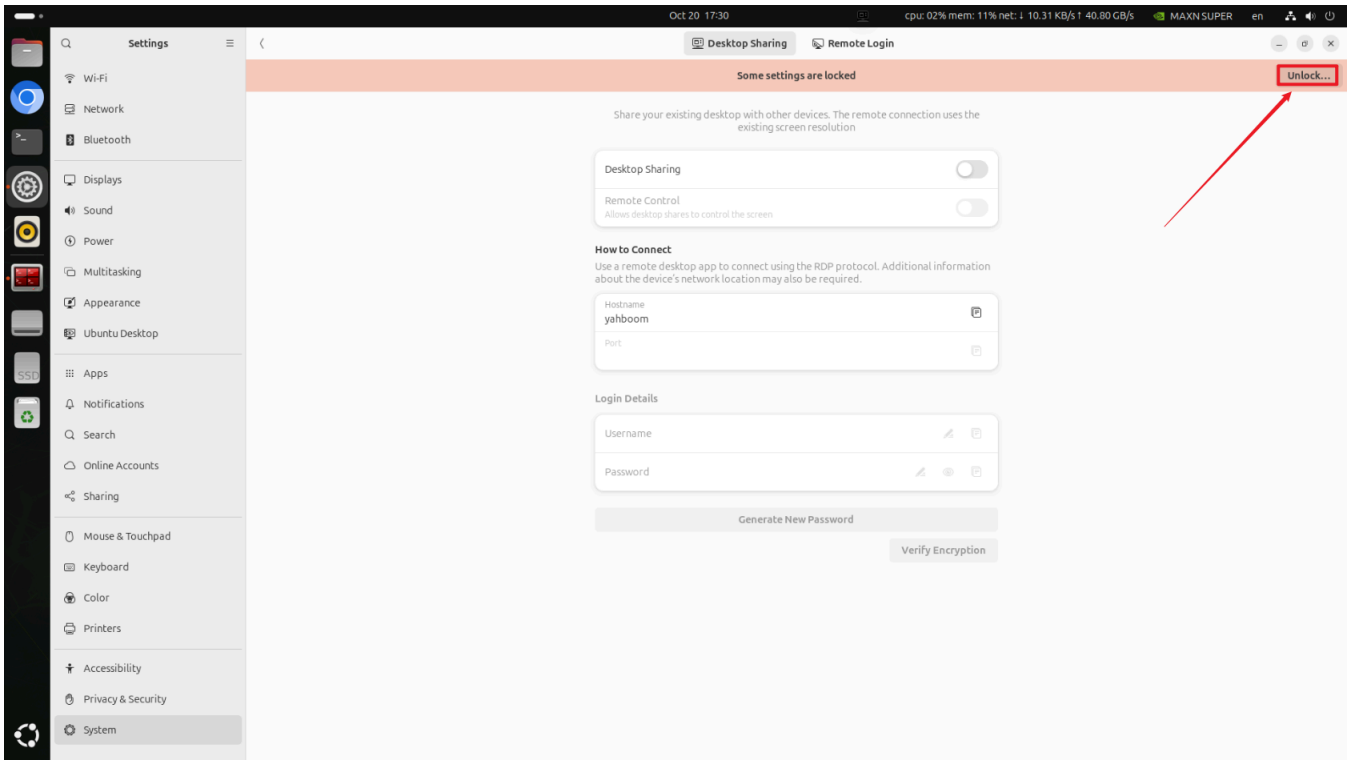
2. System settings

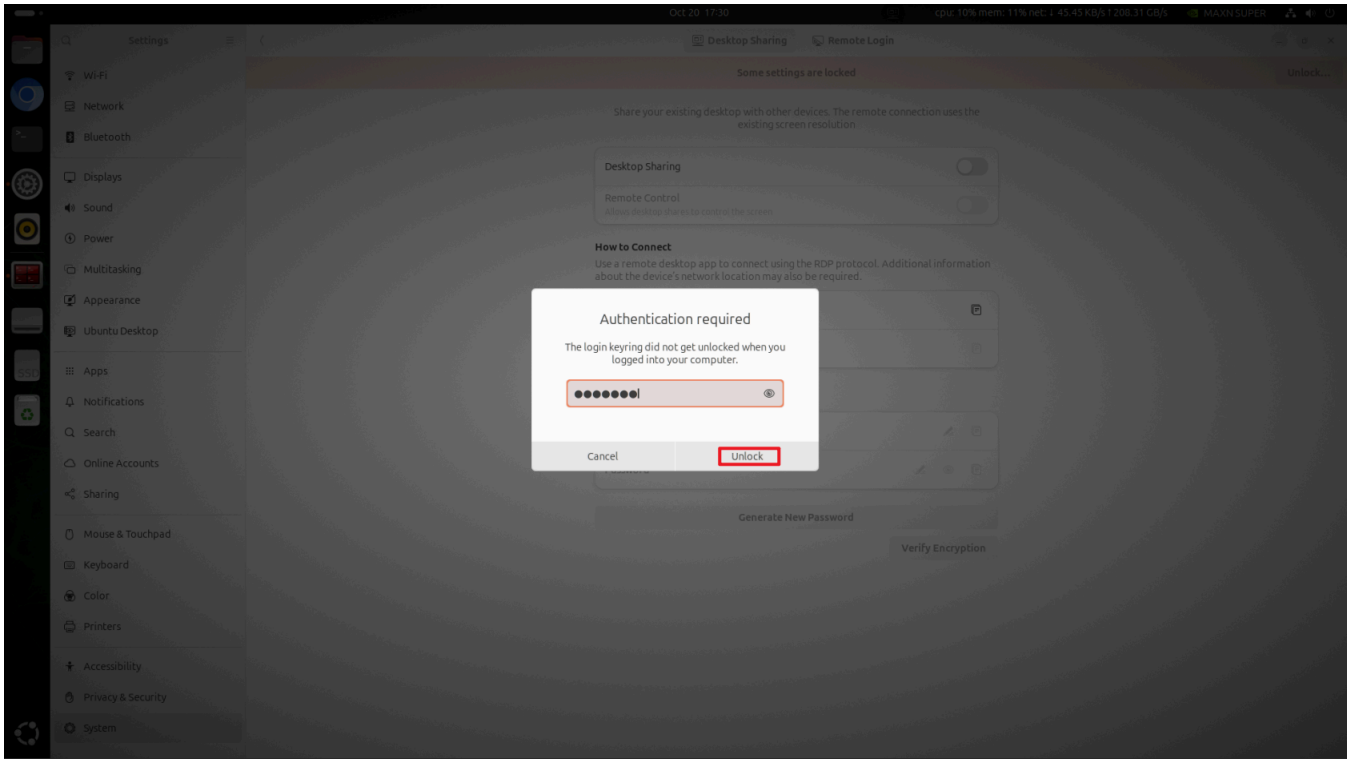
The RDP settings are located in System->Remote Desktop:

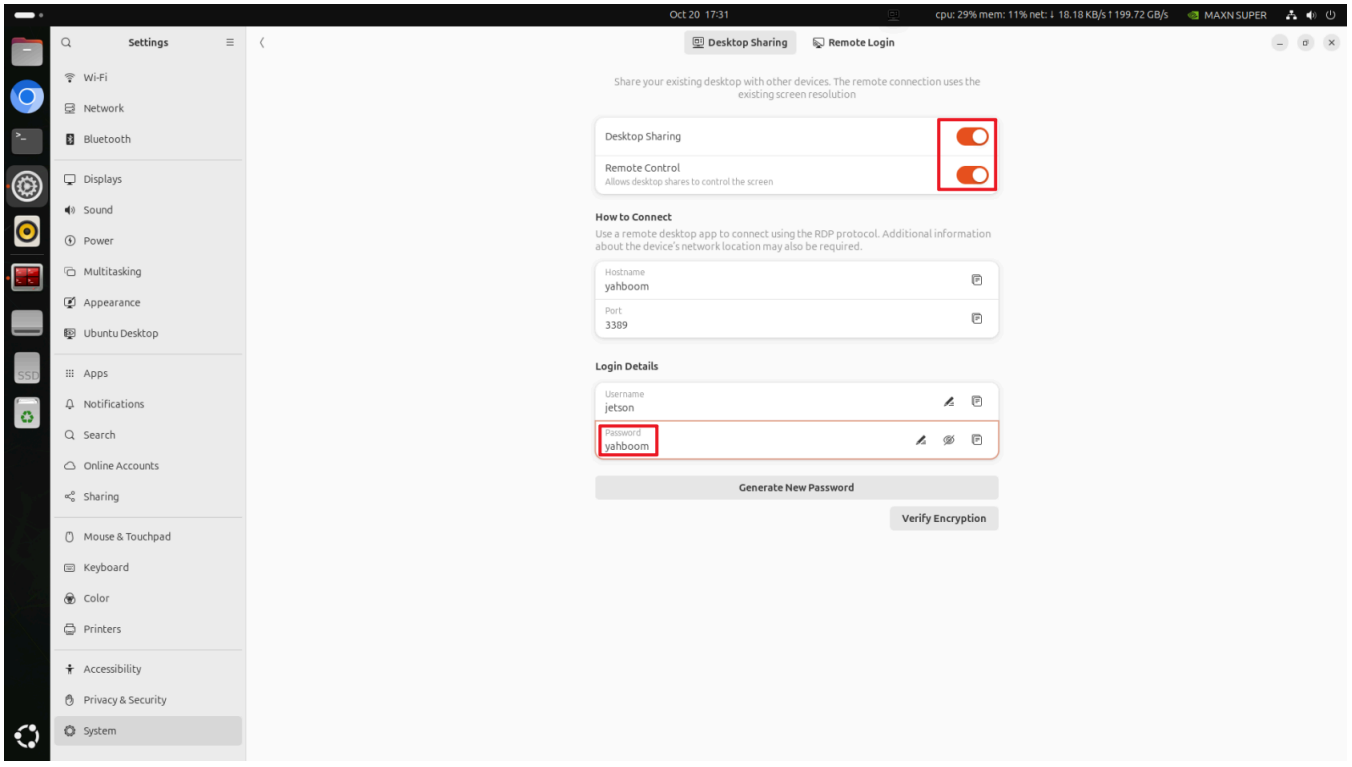


After entering it, you can freely switch between the Desktop Sharing and Remote Login interfaces. Desktop Sharing is desktop sharing, while Remote Login is remote login; the core difference lies in whether a new session is created and how it affects the current local operation.

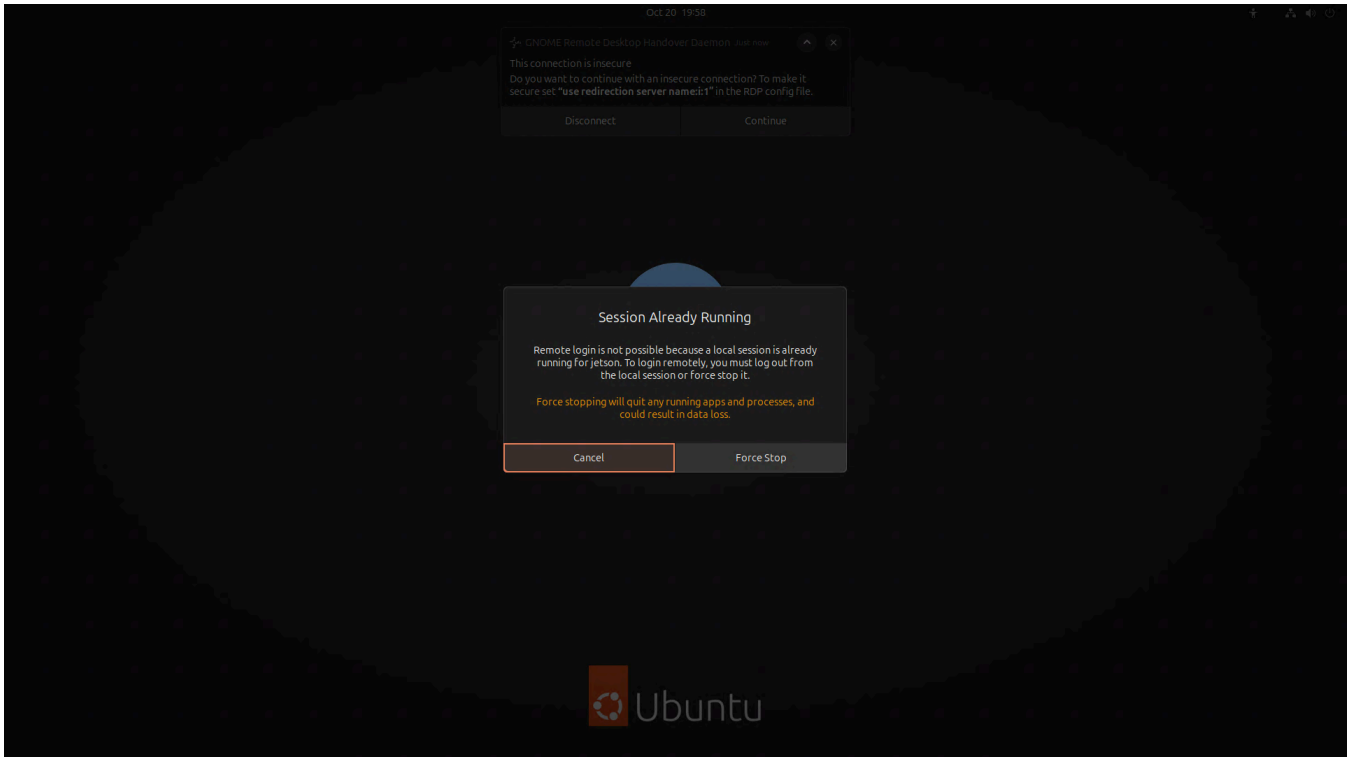
Select Desktop Sharing, click Unlock first, and enter the password to enable remote control. This method can be connected together with VNC.

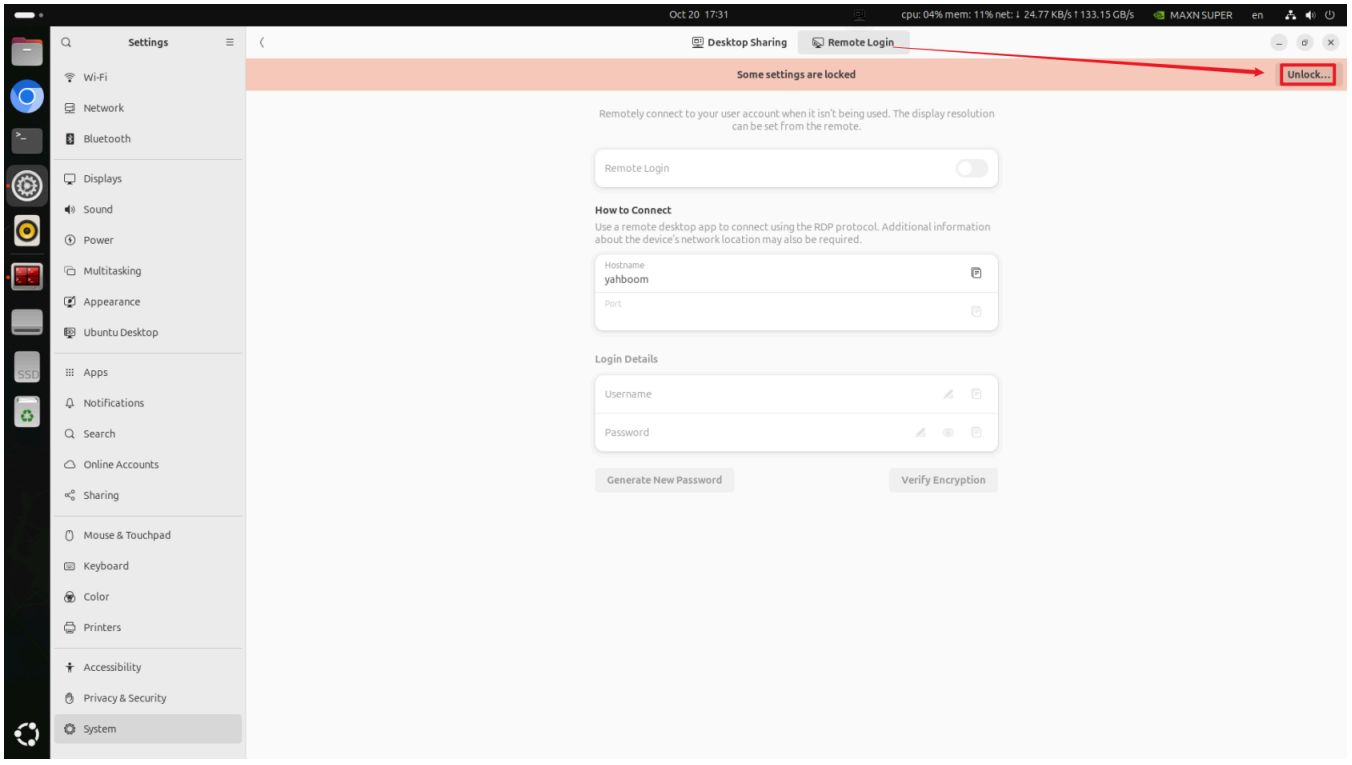


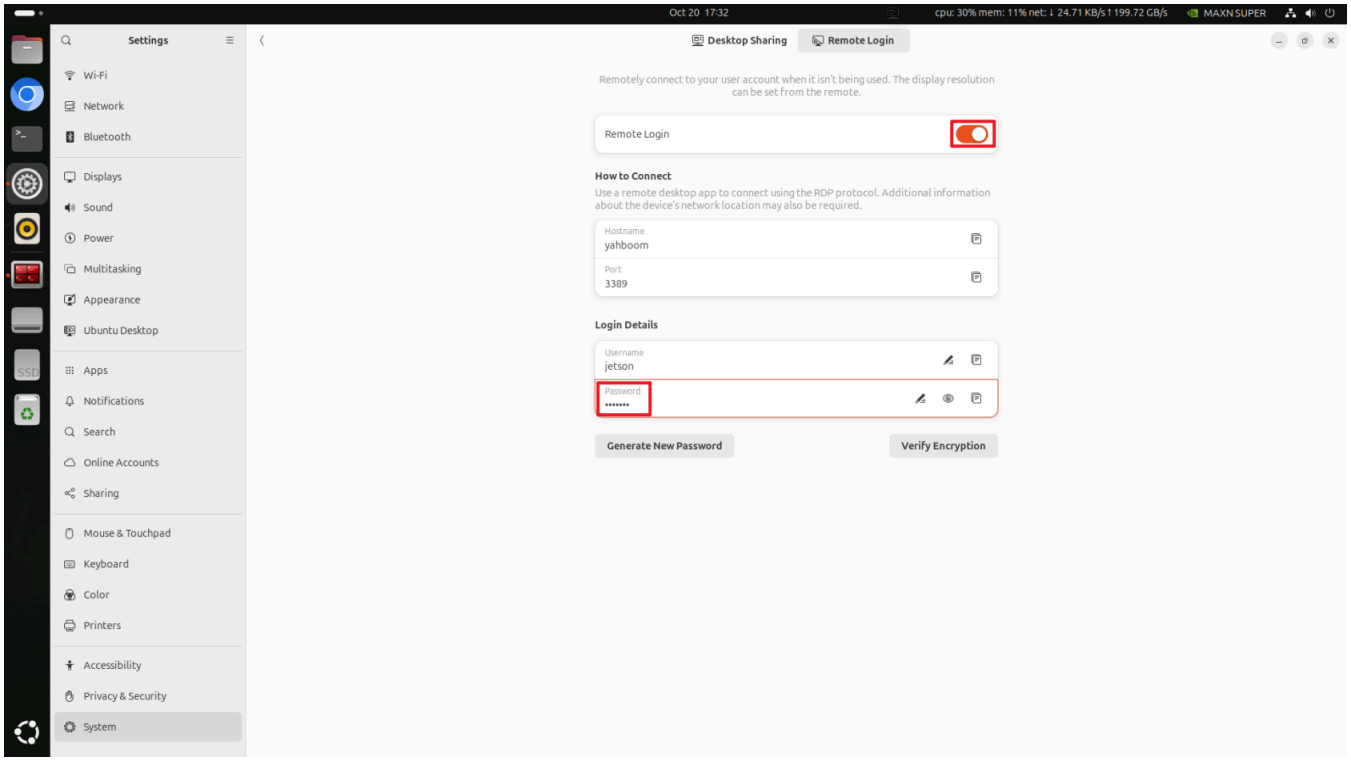




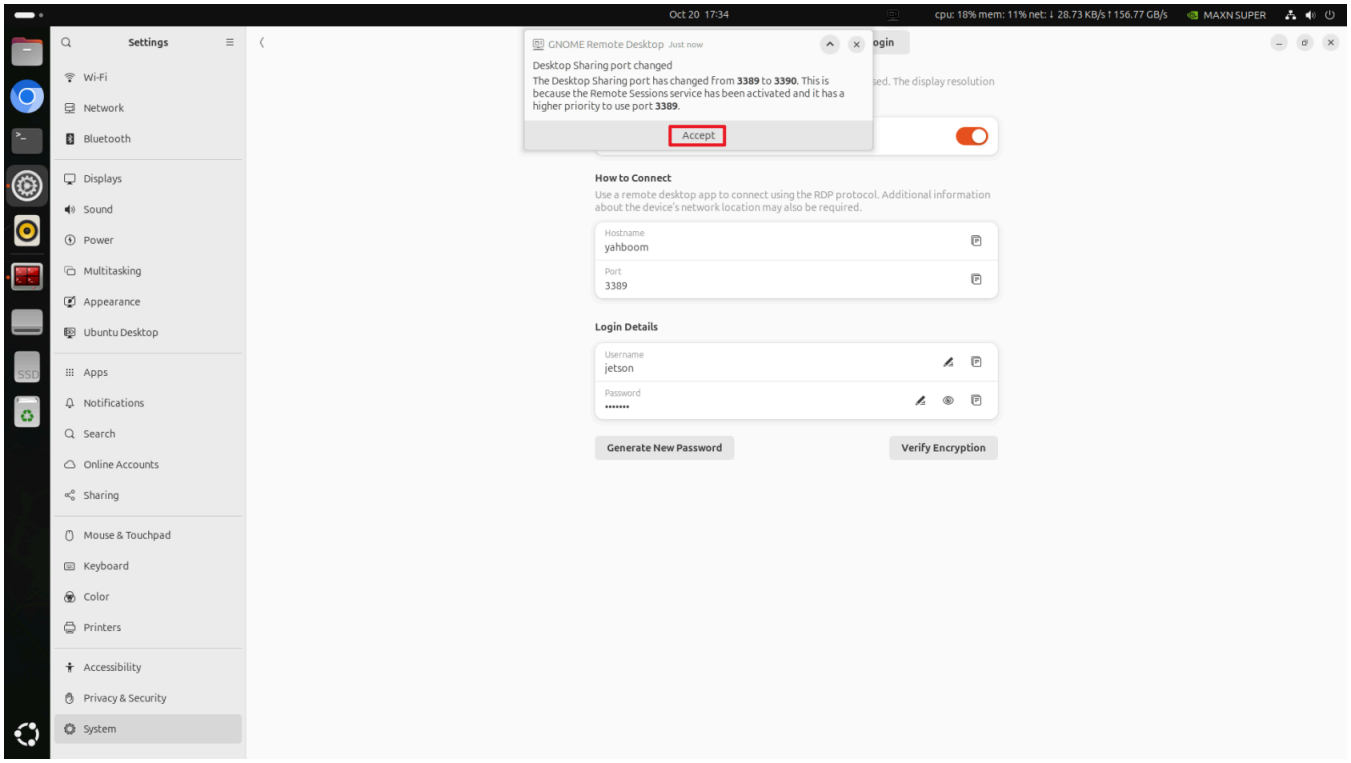
Select Remote Login, click Unlock in the same way, and enter the password to enable remote control. Because it establishes a completely new and independent virtual desktop session, VNC cannot be connected together at this time. When Windows connects to RDP with mstsc, a "Seesion Already Running" window prompt appears, and you have to switch between the VNC/RDP modes with commands.


Oct 20 19:58
GNOME Remote Desktop Handover Daemon Just now
This connection is insecure
Do you want to continue with an insecure connection? To make it secure set "use redirection server name:i:1" in the RDP config file.
Disconnect
Continue
Session Already Running
Remote login is not possible because a local session is already running for jetson. To login remotely, you must log out from the local session or force stop it.
Force stopping will quit any running apps and processes, and could result in data loss.
Cancel
Force Stop
Ubuntu


Oct 20 17:31
cpu: 04% mem: 11% net: ↓ 24.77 KB/s ↑ 133.15 GB/s
MAXN SUPER
en
Settings
Wi-Fi
Network
Bluetooth
Displays
Sound
Power
Multitasking
Appearance
Ubuntu Desktop
Apps
Notifications
Search
Online Accounts
Sharing
Mouse & Touchpad
Keyboard
Color
Printers
Accessibility
Privacy & Security
System
Desktop Sharing
Remote Login
Some settings are locked
Unlock...
Remotely connect to your user account when it isn't being used. The display resolution can be set from the remote.
Remote Login
How to Connect
Use a remote desktop app to connect using the RDP protocol. Additional information about the device's network location may also be required.
Hostname
yahboom
Port
Login Details
Username
Password
Generate New Password
Verify Encryption



> Note: after Desktop Sharing and Remote Login are enabled at the same time, the port of Desktop Sharing switches automatically to port 3390; just click Accept in the pop-up window.



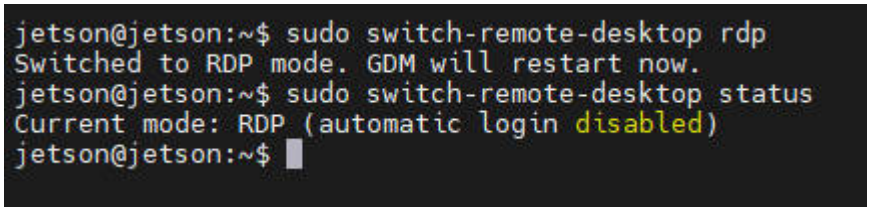
3. Switch between the VNC/RDP modes with commands

If you want to use Remote Login, that is, to use port 3389, log in to the Jetson through the terminal or over SSH and run:

```
sudo switch-remote-desktop rdp
```

This command switches to the RDP configuration and restarts `gdm3`. During the switch the local display may return to the login screen; this is normal. Check the current mode with:

```
sudo switch-remote-desktop status
```



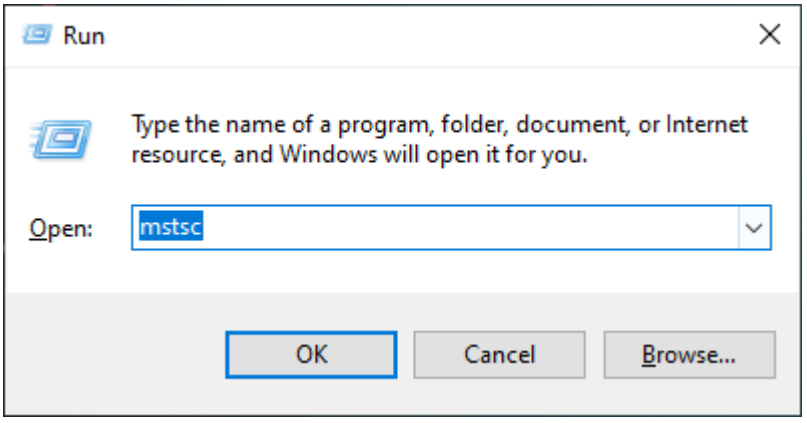
System-level RDP needs to create a graphical session for the remote user. If the Jetson has already entered the local desktop through automatic login, a local graphical session of the same user may cause the remote connection to report `Session Already Running`; switching to RDP mode turns off automatic login.

4. Connect to the Jetson from Windows

1. Check the IP address of the mainboard in the Jetson terminal:

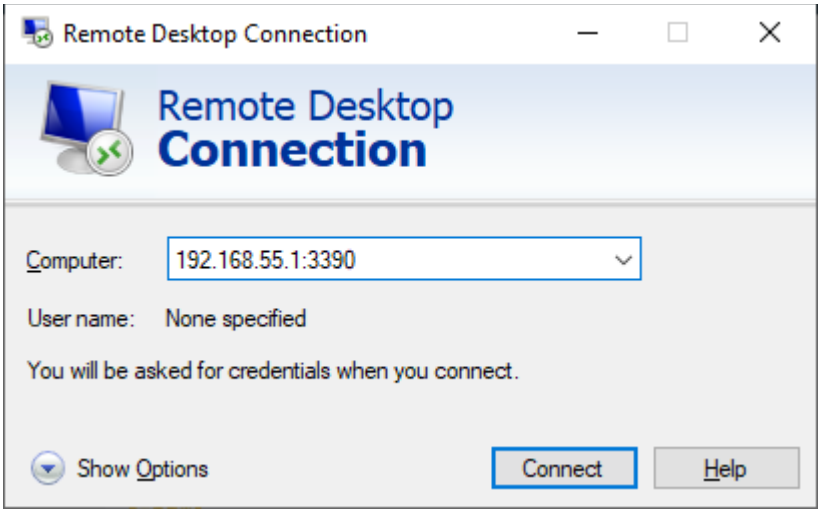
   ```
   ifconfig
   ```

2. Open "Remote Desktop Connection" on Windows (run `mstsc`).

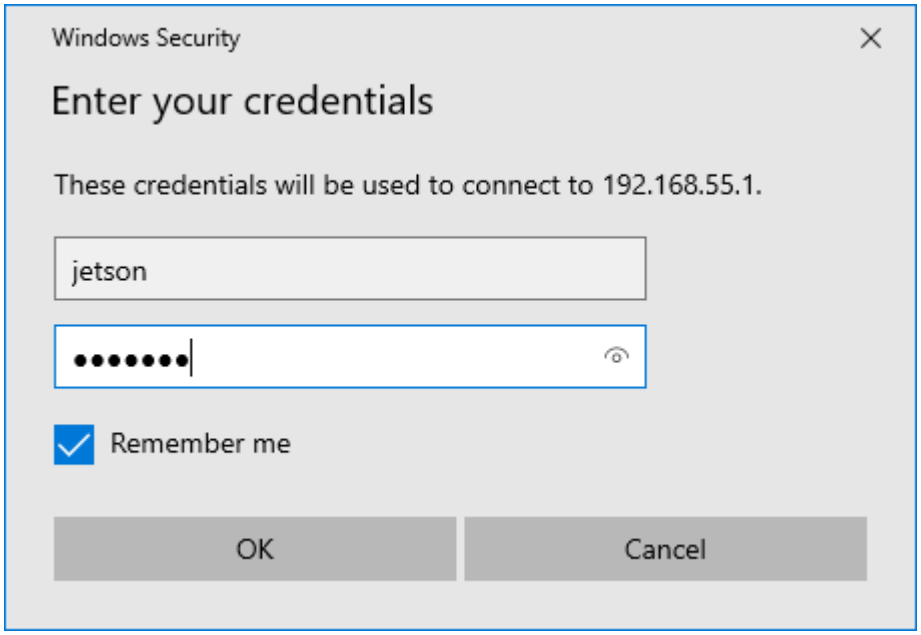


3. Enter the IP address of the Jetson under "Computer"; if you need to specify a port, use `IP:3390` or `IP:3389`, and then click "Connect".

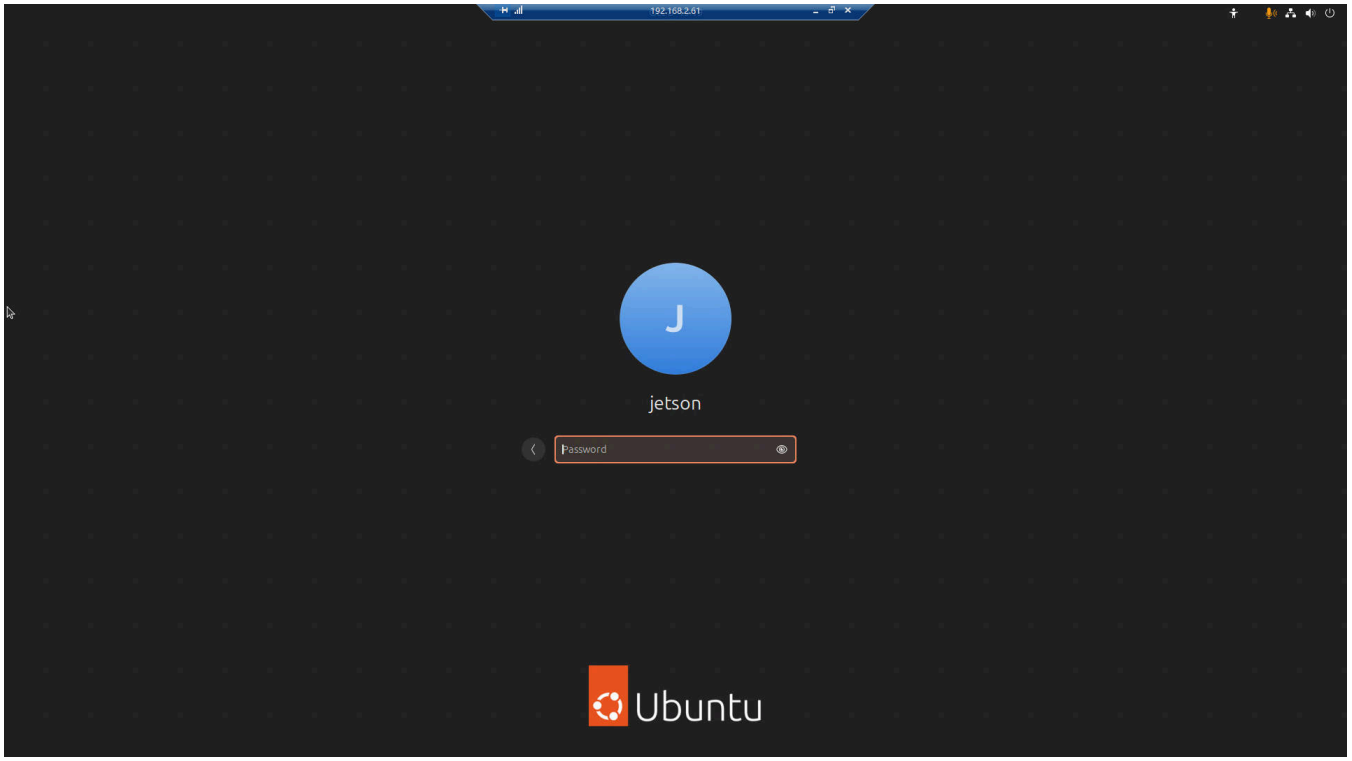
> Here 3390 is used, that is, the Desktop Sharing mode. This mode does not require running a switching command before you can log in to the rdp interface.

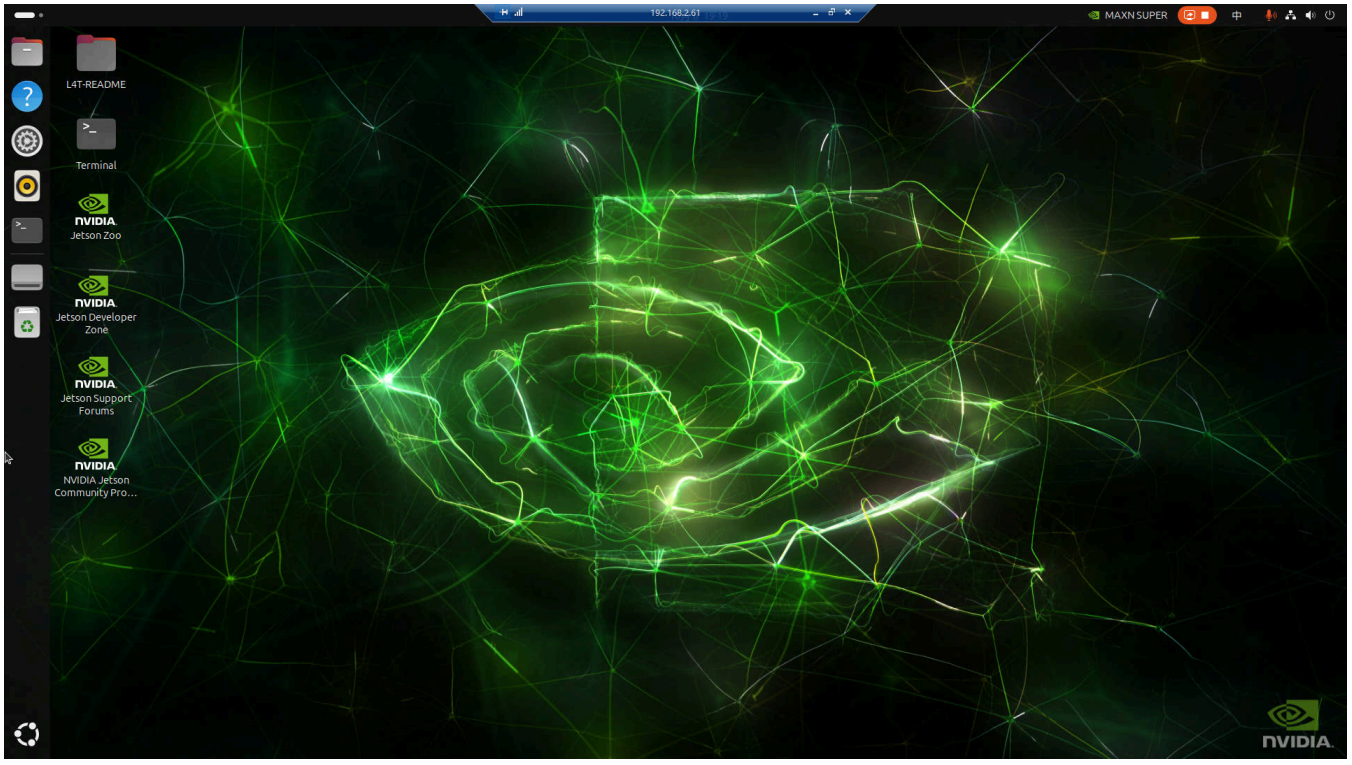


4. On the first connection, confirm the server certificate when prompted.
5. Enter the Linux user name of the Jetson and the system login password to complete the login.



After the connection succeeds, Windows shows the separate remote GNOME desktop of the Jetson.





5. Common issues

5.1. `Session Already Running` is reported

Make sure you have switched to RDP mode:

```
sudo switch-remote-desktop rdp
sudo switch-remote-desktop status
```

RDP mode requires that the remote user has no local graphical session already running. If the connection still fails, check whether the Jetson still has a local desktop session for that user and confirm that the switch command has completed.

5.2. Cannot connect or the connection times out

First confirm that Windows can reach the IP address of the Jetson, then check the service and the port on the Jetson:

```
systemctl is-active gnome-remote-desktop.service
ss -ltn | grep 3389
```

The RDP service should be running and listening on TCP `3389`. You also need to confirm that neither the network nor the firewall blocks that port.

5.3. Authentication failure

Use the Linux user name of the Jetson and the system login password, not the VNC password. If Windows has stored old credentials, delete the corresponding entry in the Windows Credential Manager and connect again.

6. Security notes

- RDP uses TCP `3389` and `3390` by default; use it only on a trusted LAN or inside a protected VPN, and do not expose it directly to the public internet.
- Keep the Jetson login password safe; never record passwords, TLS private keys or certificate contents in tutorials, screenshots or public repositories.
- Switching the mode restarts the graphical login manager and interrupts the current graphical session; confirm that a fallback entry such as SSH is available before operating.